Phase 0 report – Stochastic-LI&F bridge & calibration

Verdict: **PASS**

File inventory

The following modules were created under `./deq/closed_form/`:

- `stochastic_lif.py` – N-neuron oracle wrapping `../archetypes/lif_fcs.py:simulate` with per-copy threshold heterogeneity and Bernoulli per-tick input thinning. Returns population-mean firing rates and per-copy spike trains.
- `phase0_infrastructure.py` – this validation script.
- `README.md`, `make_all.sh`, `.gitignore` – workspace boilerplate.

Validation V0.1 – Population f-I curve

Setup: single uncoupled neuron, $N = 100$ copies, threshold jitter $\varepsilon = 2$, sweep external drive weight $w_X \in \{2, 4, ..., 20\}$, $T = 1000$ ticks. Three thinning levels $p_{\text{thin}} \in \{1.0, 0.7, 0.4\}$.

Monotonicity per curve (drive vs rate non-decreasing): p=1.0: yes, p=0.7: yes, p=0.4: yes.

Saturation maxima per curve: p=1.0: 1.000, p=0.7: 0.701, p=0.4: 0.401.

Acceptance: each curve monotone, $p = 1.0$ saturating ≥ 0.4, smaller p_{thin} gives strictly lower saturation. **PASS**.

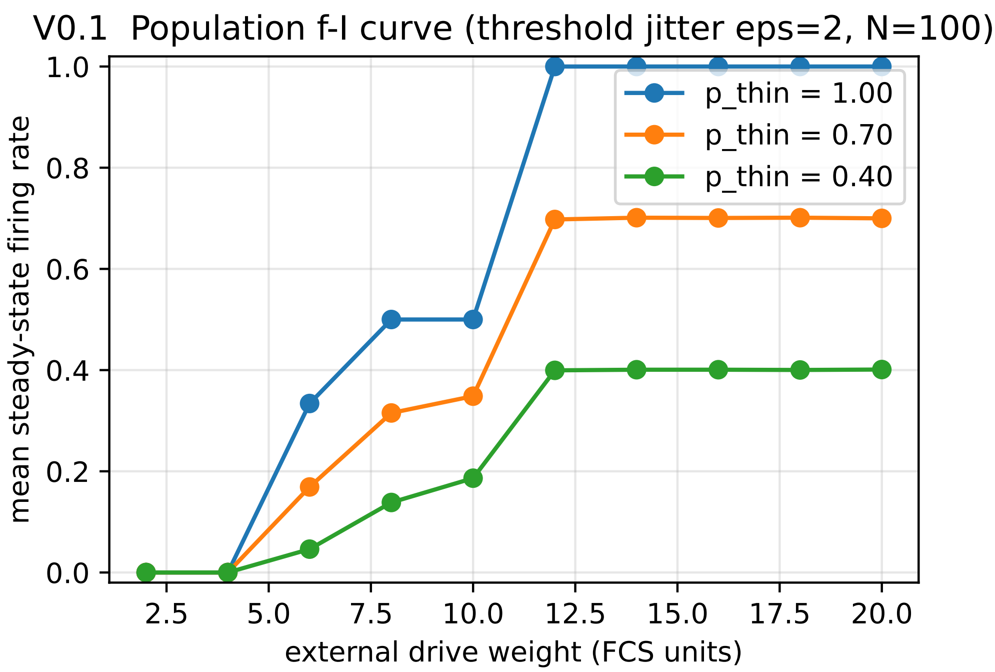

Figure 1: V0.1: population f-I curve at three thinning levels. The $p_{\text{thin}} = 1.0$ curve is the deterministic-input case (variance from threshold jitter only); lower p_{thin} injects Bernoulli input variance, which both reduces the mean drive and reshapes the curve.

Validation V0.2 – ISI CV calibration

Setup: single neuron at drive = 11 (FCS default), $N = 100$ copies, $T = 2000$ ticks. Sweep $p_{\text{thin}} \in \{1.0, 0.85, 0.7, 0.5, 0.3\}$ and threshold jitter $\varepsilon \in \{0, 2, 5, 10\}$.

Maximum per-cell wallclock: 0.0 s (threshold 30 s).

Recommended operating point: p_thin = 0.70, jitter = 0, CV = 0.547, rate = 0.701.

Number of $(p_{\text{thin}}, \varepsilon)$ combinations with CV ≥ 0.5: 12 of 20.

Acceptance: at least one combination achieves CV ≥ 0.5 and total runtime fits in budget. **PASS**.

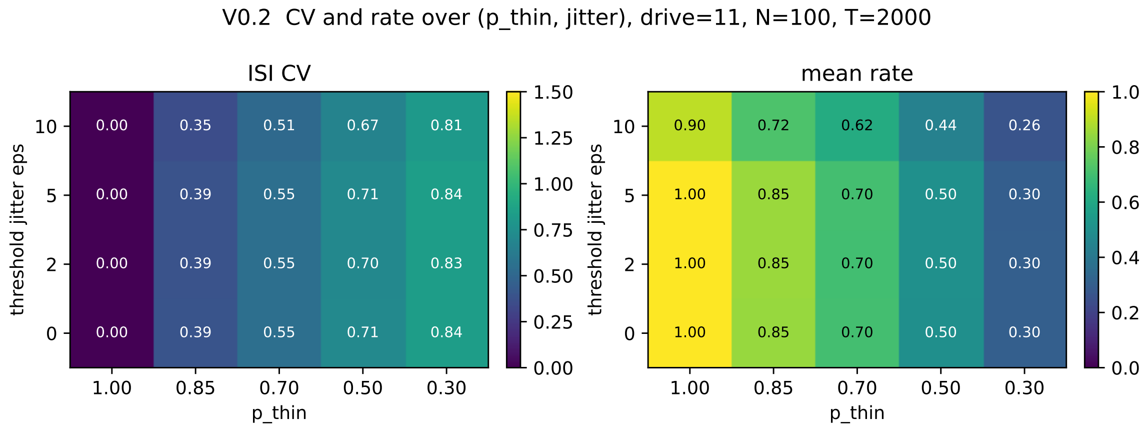

Figure 2: V0.2: ISI coefficient of variation (left) and mean firing rate (right) across the $(p_{\text{thin}}, \varepsilon)$ grid. CV ≈ 1 corresponds to Poisson-like firing where Siegert's diffusion approximation is valid; CV $\ll 0.5$ means quasi-deterministic firing where the diffusion approximation underestimates regularity.

Overall verdict

PASS.

The recommended operating point (p_thin = 0.70, jitter = 0, CV = 0.547, rate = 0.701) is locked in for Phase 1's Siegert comparison. The Bernoulli-thinning + threshold-jitter combination produces input statistics in the diffusion-approximation-valid regime without breaking the FCS-LI&F oracle's per-tick semantics.